

STANDARD 2

Stage 6

Smarter Maths

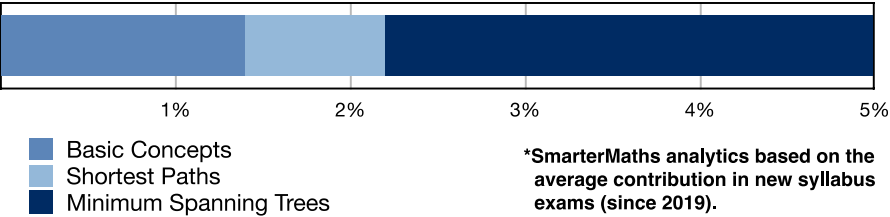
Networks (Std 2), N2 Network Concepts (Y12)

Minimum Spanning Trees

Teacher: Hailey Summer

Exam Equivalent Time: 99 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History N2 Network Concepts



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

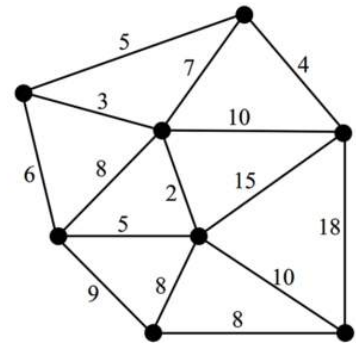
- *MS-N2 Network Concepts* has contributed 5.0% to new syllabus exams since its introduction in 2019.
- We have split this area into three categories for the purposes of analysis: 1-*Basic Concepts* (1.4%), 2-*Shortest Paths* (0.8%) and 3-*Minimum Spanning Trees* (2.8%).
- This analysis will look at the largest contributing sub-topic, *Minimum Spanning Trees* (2.8%).

ANALYSIS - What to Expect and Common pitfalls

- *Minimum Spanning Trees* (2.8%) looks to be an exam staple after appearing in every new syllabus exam to date with meaningful allocations of 2-4 marks.
- The database reflects this importance and besides the growing number of HSC past questions, includes many supplementary examples informed by the syllabus document, NESA's Networks Topic Guide, exemplar questions and past VCE exam questions.
- *Minimum Spanning Trees* represents a high scoring opportunity for students. We expect it to regularly be tested with typical levels of difficulty around the Band 3-4 level, although *2023 Std2 19b* produced a mean mark of just 46% and deserves attention.
- The worked solutions provide significant detail for solving problems using both Prim's and Kruskal's algorithms, reflecting the step by step approach outlined in NESA's Networks Topic Guidance document.

Questions

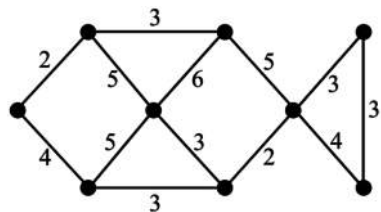
1. Networks, STD2 N2 2010 FUR1 5 MC



For the network above, the length of the minimal spanning tree is

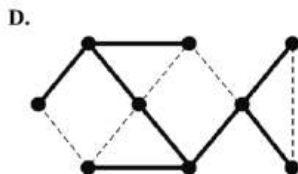
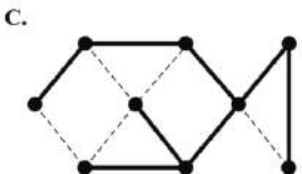
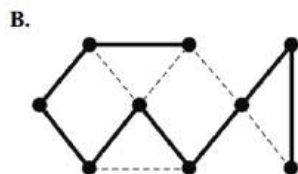
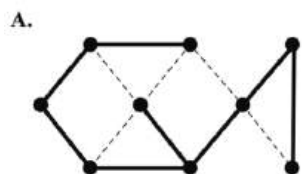
- A. 30
- B. 31
- C. 35
- D. 39

2. Networks, STD2 N2 2013 FUR1 3 MC

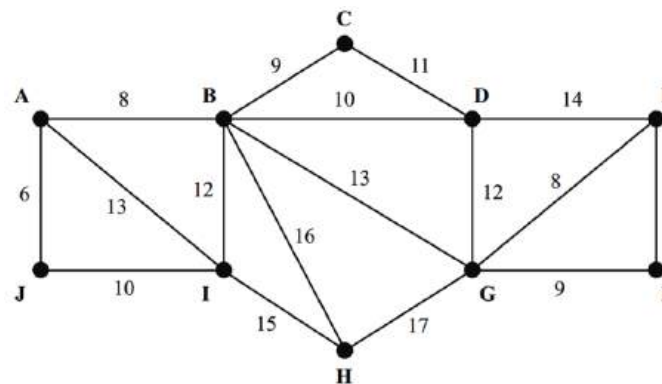


The vertices of the graph above represent nine computers in a building. The computers are to be connected with optical fibre cables, which are represented by edges. The numbers on the edges show the costs, in hundreds of dollars, of linking these computers with optical fibre cables.

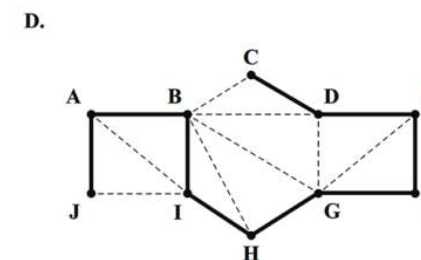
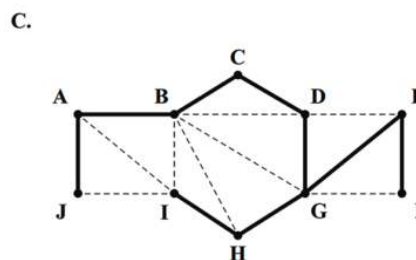
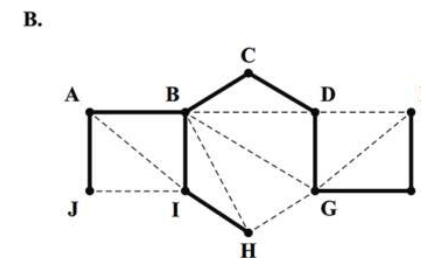
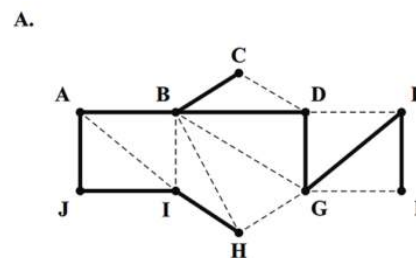
Based on the same set of vertices and edges, which one of the following graphs shows the cable layout (in bold) that would link all the computers with optical fibre cables for the minimum cost?



3. Networks, STD2 N2 2014 FUR1 5 MC

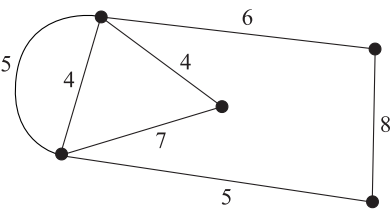


Which one of the following is the minimal spanning tree for the weighted graph shown above?



4. Networks, STD2 N2 SM-Bank 26 MC

A weighted network diagram is shown below.

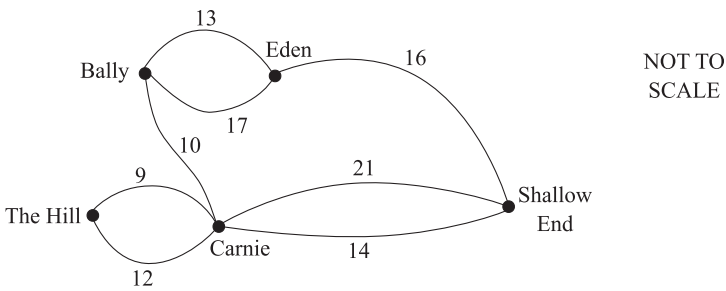


What is the weight of the minimum spanning tree?

- A. 18
- B. 19
- C. 20
- D. 22

5. Networks, STD2 N2 SM-Bank 27 MC

This diagram shows the possible paths (in km) for laying gas pipes between various locations.

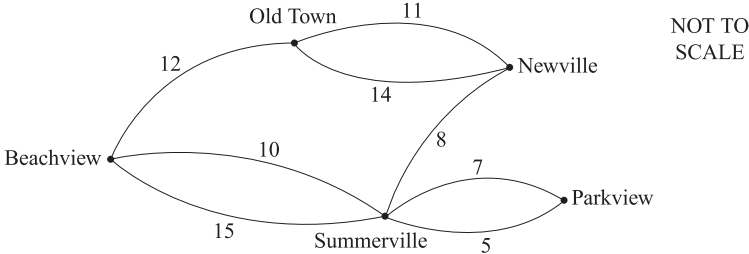


Gas is be supplied from one location. Any one of the locations can be the source of the supply.
What is the minimum total length of length of the pipes required to provide gas to all the locations?

- A. 46 km
- B. 48 km
- C. 50 km
- D. 52 km

6. Networks, STD2 N2 SM-Bank 1 MC

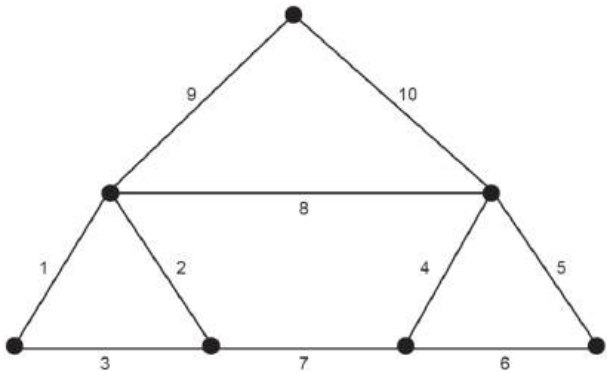
This diagram shows the possible paths (in km) for laying gas pipes between various locations.



Gas is to be supplied from one location. Any one of the locations can be the source of the supply.
What is the minimum total length of the pipes required to provide gas to all the locations?

- A. 32 km
- B. 34 km
- C. 36 km
- D. 38 km

7. Networks, STD2 N2 2006 FUR1 4 MC

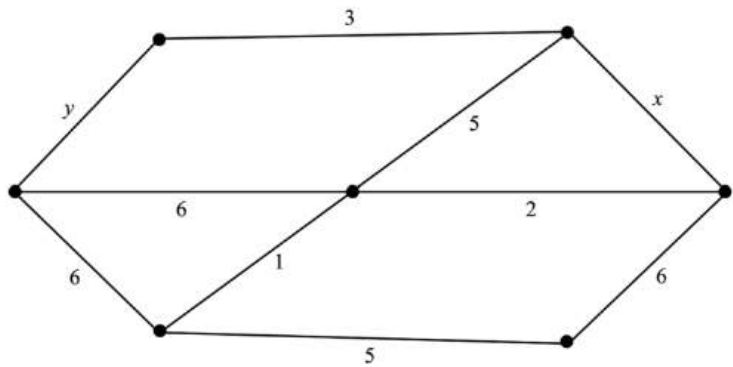


The minimal spanning tree for the network above will include the edge that has a weight of

- A. 3
- B. 6
- C. 8
- D. 9

8. Networks, STD2 N2 2007 FUR1 7 MC

The minimal spanning tree for the network below includes two edges with weightings x and y .



The length of the minimal spanning tree is 19.

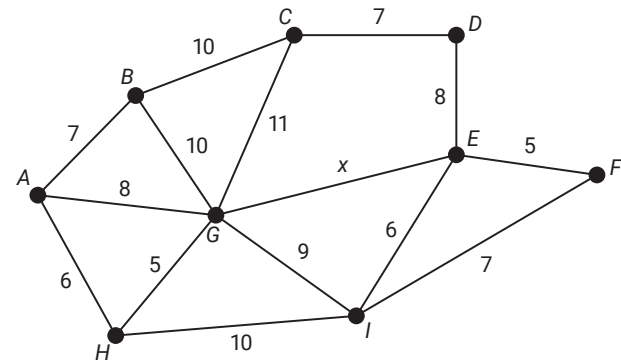
The values of x and y could be

- A. $x = 1$ and $y = 7$
- B. $x = 2$ and $y = 5$
- C. $x = 3$ and $y = 5$
- D. $x = 4$ and $y = 5$

9. Networks, STD2 N2 2020 FUR1 5 MC

The network below shows the distances, in metres, between camp sites at a camping ground that has electricity.

The vertices A to I represent the camp sites.



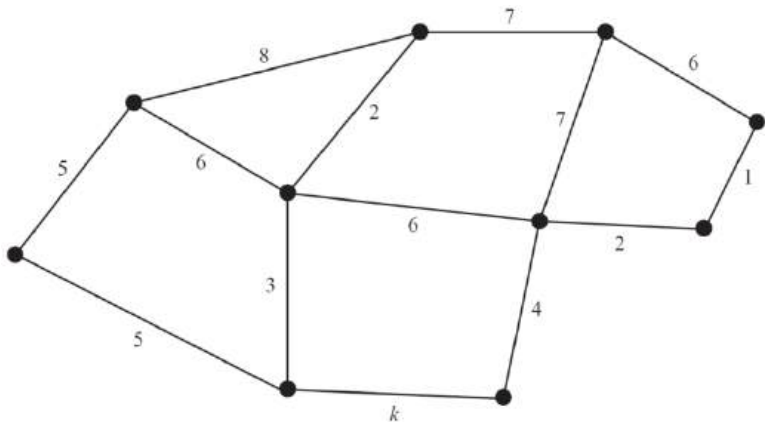
The minimum length of cable required to connect all the camp sites is 53 m.

The value of x , in metres, is at least

- A. 6
- B. 8
- C. 9
- D. 11

10. Networks, STD2 N2 2016 FUR1 4 MC

The minimum spanning tree for the network below includes the edge with weight labelled k .



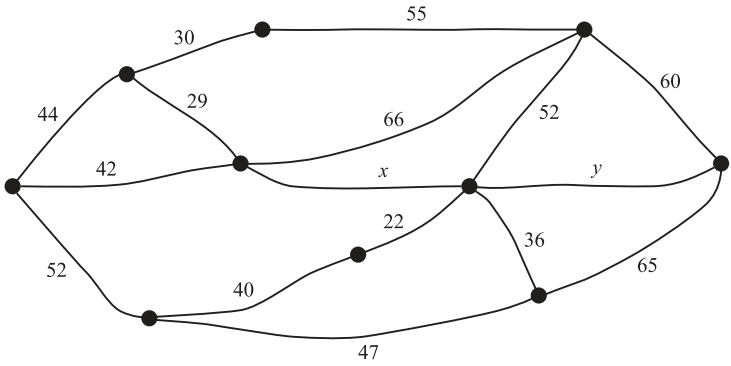
The total weight of all edges for the minimum spanning tree is 33.

The value of k is

- A. 2
- B. 3
- C. 4
- D. 5

11. Networks, STD2 N2 2021 FUR1 8 MC

A network of roads connecting towns in an alpine region is shown below.
The distances between neighbouring towns, represented by vertices, are given in kilometres.



The region receives a large snowfall, leaving all roads between the towns closed to traffic.
To ensure each town is accessible by car from every other town, some roads will be cleared.
The minimal total length of road, in kilometres, that needs to be cleared is

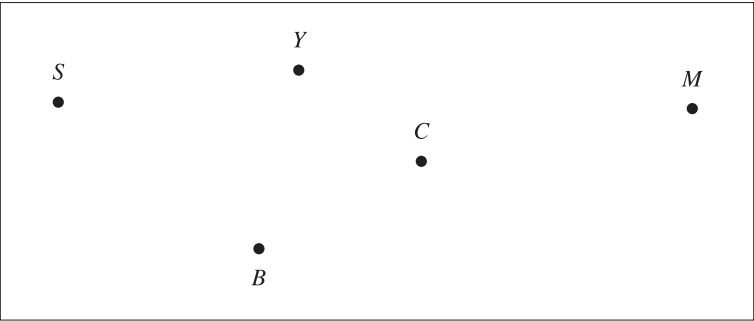
- A. 361 if $x = 50$ and $y = 55$
- B. 361 if $x = 50$ and $y = 60$
- C. 366 if $x = 55$ and $y = 55$
- D. 371 if $x = 55$ and $y = 65$

12. Networks, STD2 N2 2022 HSC 20

The table below shows the distances, in kilometres, between a number of towns.

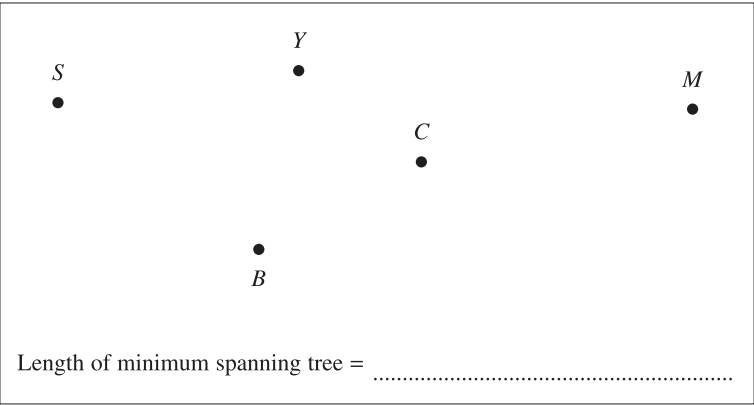
Towns	Snowtown (S)	Clairville (C)	Yuma (Y)	Bosten (B)	Morrella (M)
(S)	–	–	280	275	–
(C)	–	–	60	150	–
(Y)	280	60	–	–	530
(B)	275	150	–	–	790
(M)	–	–	530	790	–

a. Using the vertices given, draw a weighted network diagram to represent the information shown in the table. (2 marks)



b. A tourist wishes to visit each town.

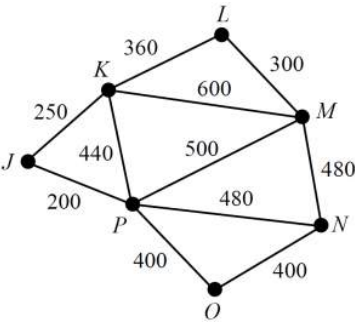
Draw the minimum spanning tree which will allow for this AND determine its length. (3 marks)



13. Networks, STD2 N2 2015 FUR2 1

A factory requires seven computer servers to communicate with each other through a connected network of cables.

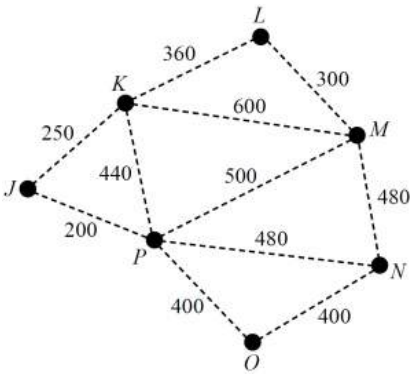
The servers, **J**, **K**, **L**, **M**, **N**, **O** and **P**, are shown as vertices on the graph below.



The edges on the graph represent the cables that could connect adjacent computer servers.

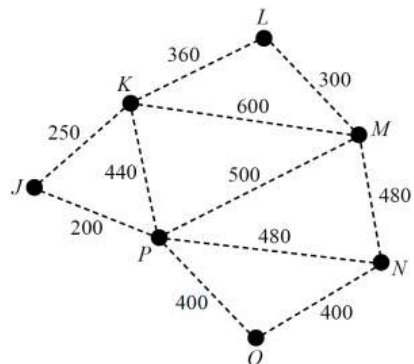
The numbers on the edges show the cost, in dollars, of installing each cable.

- a. What is the cost, in dollars, of installing the cable between server **L** and server **M**? (1 mark)
- b. What is the cheapest cost, in dollars, of installing cables between server **K** and server **N**? (1 mark)
- c. The computer servers will be able to communicate with all the other servers as long as each server is connected by cable to at least one other server.
 - i. The cheapest installation that will join the seven computer servers by cable in a connected network follows a minimum spanning tree.
Draw the minimum spanning tree on the plan below. (1 mark)



- ii. The factory's manager has decided that only six connected computer servers will be needed, rather than seven.
How much would be saved in installation costs if the factory removed computer server **P** from its minimum spanning tree network?

A copy of the graph above is provided below to assist with your working. (1 mark)



14. Networks, STD2 N2 2008 FUR2 1

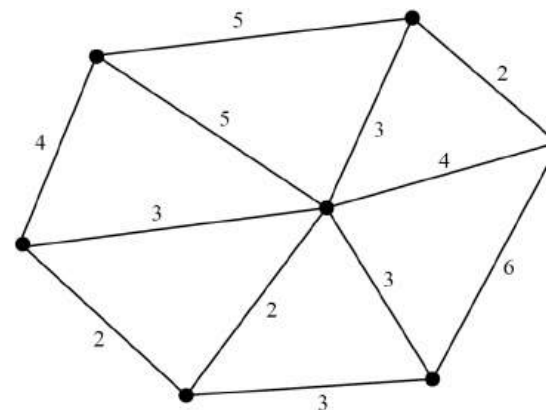
James, Dante, Tahlia and Chanel are four children playing a game.

In this children's game, seven posts are placed in the ground.

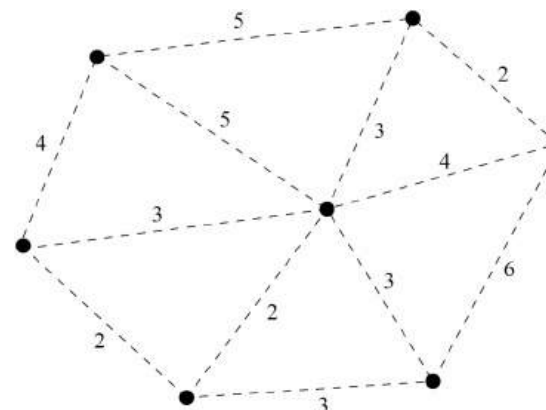
The network below shows distances, in metres, between the seven posts.

The aim of the game is to connect the posts with ribbon using the shortest length of ribbon.

This will be a minimal spanning tree.



a. Draw in a minimal spanning tree for this network on the diagram below. (1 mark)

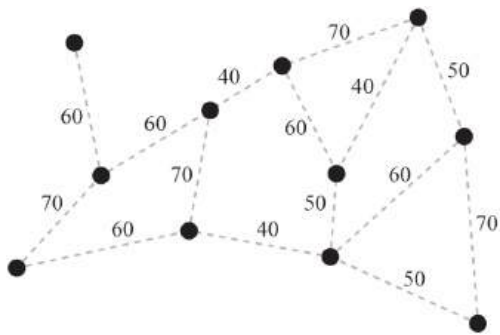


b. Determine the length, in metres, of this minimal spanning tree. (1 mark)

c. How many different minimal spanning trees can be drawn for this network? (1 mark)

15. Networks, STD2 N2 2011 FUR2 2

At the Farnham showgrounds, eleven locations require access to water. These locations are represented by vertices on the network diagram shown below. The dashed lines on the network diagram represent possible water pipe connections between adjacent locations. The numbers on the dashed lines show the minimum length of pipe required to connect these locations in metres.

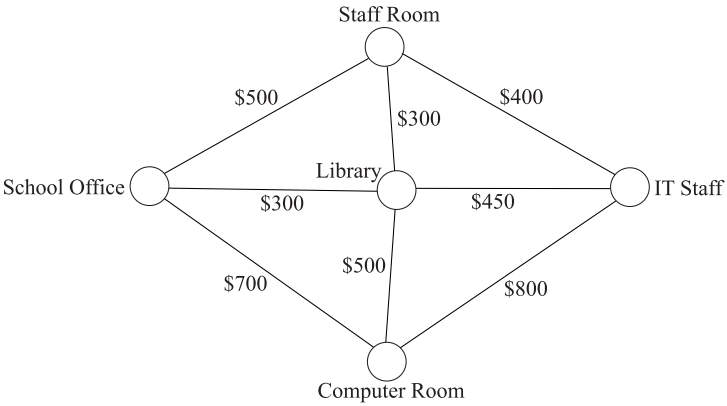


All locations are to be connected using the smallest total length of water pipe possible.

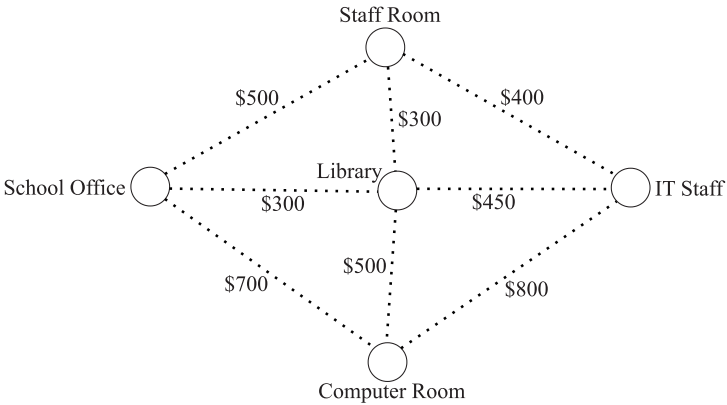
- a. On the diagram, show where these water pipes will be placed. (1 mark)
- b. Calculate the total length, in metres, of water pipe that is required. (1 mark)

16. Networks, STD2 N2 SM-Bank 2

A school is designing a computer network between five key areas within the school. The cost of connecting the rooms is shown in the diagram below.



- i. Complete the spanning tree below that creates the school's network at a minimum cost. (1 mark)



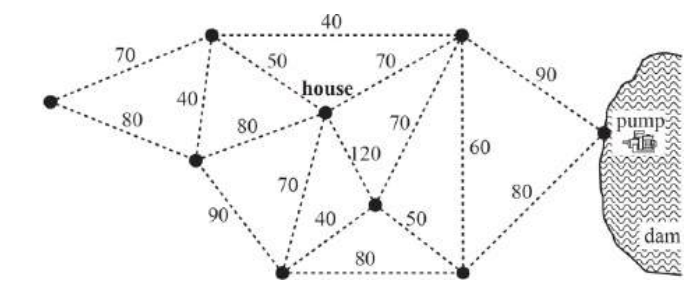
- ii. What is the minimum cost of the network? (1 mark)

17. Networks, STD2 N2 SM-Bank 14

Water will be pumped from a dam to eight locations on a farm.

The pump and the eight locations (including the house) are shown as vertices in the network diagram below.

The numbers on the edges joining the vertices give the shortest distances, in metres, between locations.



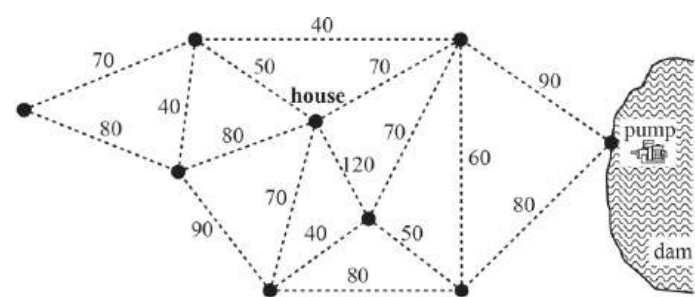
i. How many vertices on the network diagram have an odd degree? (1 mark)

The total length of all edges in the network is 1180 metres.

The total length of pipe that supplies water from the pump to the eight locations on the farm is a minimum.

This minimum length of pipe is laid along some of the edges in the network.

ii. On the diagram below, **draw** the minimum length of pipe that is needed to supply water to all locations on the farm. (2 marks)



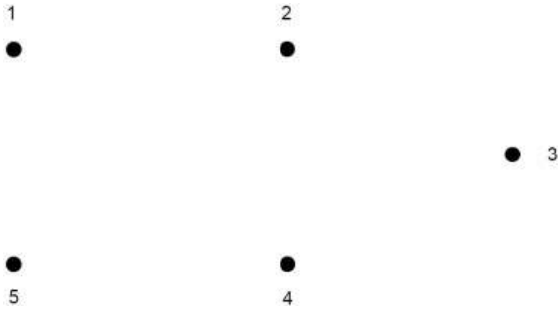
iii. What is the mathematical term that is used to describe this minimum length of pipe? (1 mark)

18. Networks, STD2 N2 2007 FUR2 1

A new housing estate is being developed.

There are five houses under construction in one location.

These houses are numbered as points 1 to 5 below.

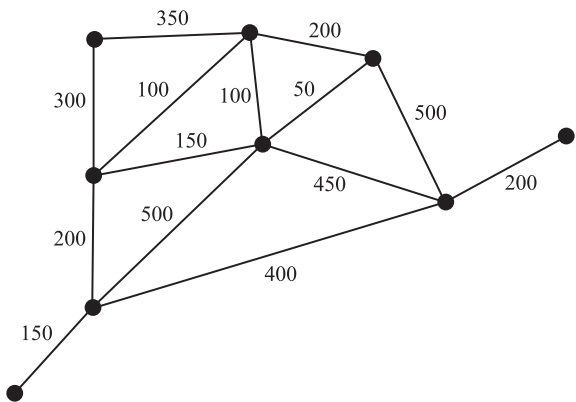


The builders require the five houses to be connected by electrical cables to enable the workers to have a supply of power on each site.

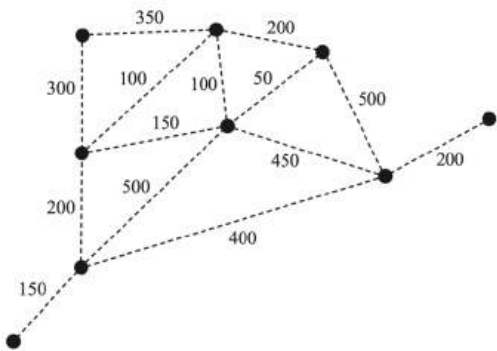
- a. What is the **minimum** number of edges needed to connect the five houses? (1 mark)
- b. On the diagram above, **draw** a connected graph with this number of edges. (1 mark)

19. Networks, STD2 N2 2017 FUR2 3a

While on holiday, four friends visit a theme park where there are nine rides.
On the graph below, the positions of the rides are indicated by the vertices.
The numbers on the edges represent the distances, in metres, between rides.

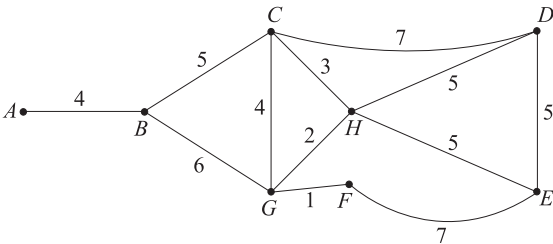


- Electrical cables are required to power the rides.
These cables will form a connected graph.
The shortest total length of cable will be used.
- a. Give a mathematical term to describe a graph that represents these cables. (1 mark)
 - b. Draw in the graph that represents these cables on the diagram below. (1 mark)



20. Networks, STD2 N2 2019 HSC 30

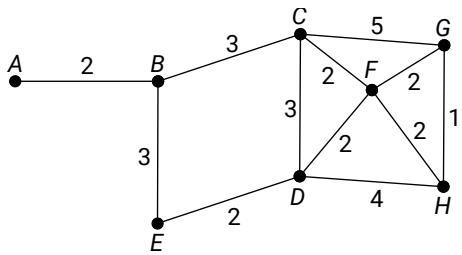
The network diagram shows the tracks connecting 8 picnic sites in a nature park. The vertices **A** to **H** represents the picnic sites. The weights on the edges represent the distance along the tracks between the picnic sites, in kilometres.



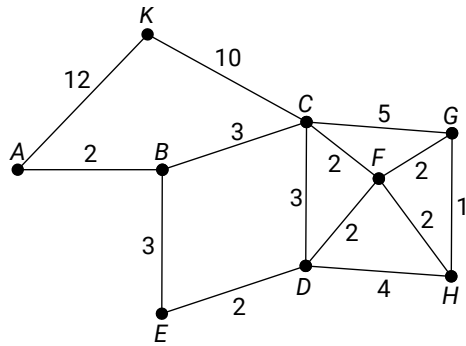
- a. Each picnic site needs to provide drinking water. The main water source is at site **A**.
Draw a minimum spanning tree and calculate the minimum length of water pipes required to supply water to all the sites if the water pipes can only be laid along the tracks. (2 marks)
- b. One day, the track between **C** and **H** is closed. State the vertices that identify the shortest path from **C** to **E** that avoids the closed track. (1 mark)

21. Networks, STD2 N2 2020 HSC 18

The diagram represents a network with weighted edges.



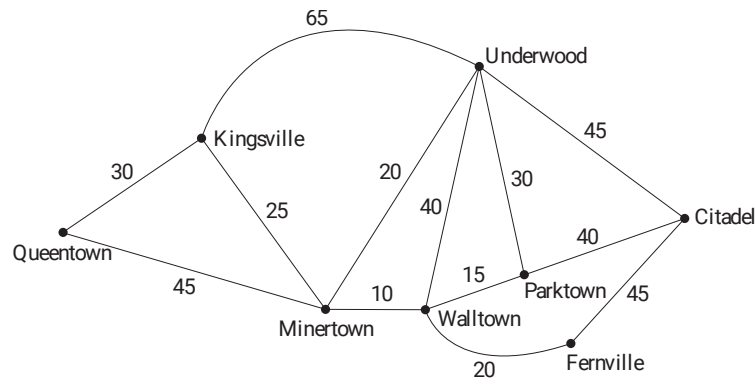
- a. Draw a minimum spanning tree for this network and determine its length. (3 marks)
- b. The network is revised by adding another vertex, **K**. Edges **AK** and **CK** have weights of 12 and 10 respectively, as shown.



What is the length of the minimum spanning tree for this revised network? (1 mark)

22. Networks, STD2 N2 2021 HSC 23

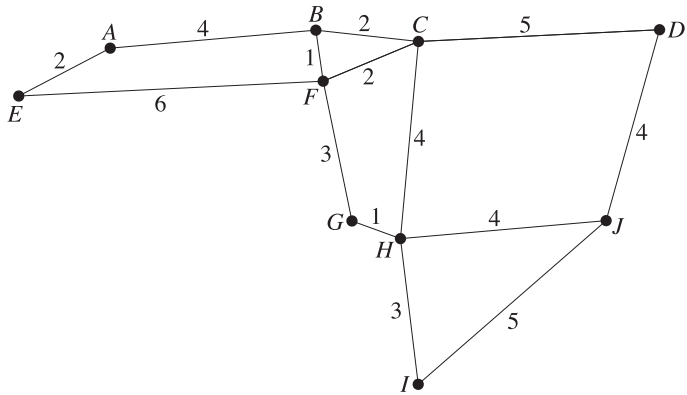
The network diagram shows the travel times in minutes along roads connecting a number of different towns.



- a. Draw a minimum spanning tree for this network and determine its length. (3 marks)
- b. How long does it take to travel from Queentown to Underwood using the fastest route? (1 mark)

23. Networks, STD2 N2 2024 HSC 18

The diagram shows a network with weighted edges.



a. Draw a minimum spanning tree for this network and determine its weight. (2 marks)

A

B

C

D

E

F

G

H

I

J

Weight =

b. Is it possible to find another spanning tree with the same weight? Give a reason for your answer. (1 mark)

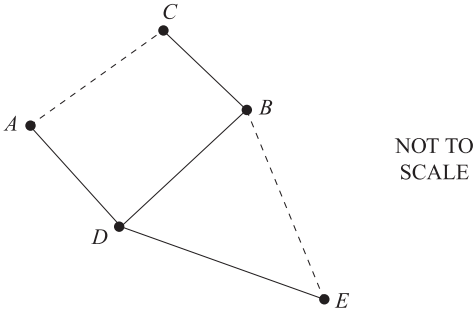
24. Networks, STD2 N2 SM-Bank 30

An amusement park has five toilet areas, **A**, **B**, **C**, **D** and **E**, which are connected by pathways.

The table shows the length of some of the pathways, in metres.

	A	B	C	D	E
A	–	–	?	450	–
B	–	–	200	300	?
C	?	200	–	–	–
D	450	300	–	–	500
E	–	?	–	500	–

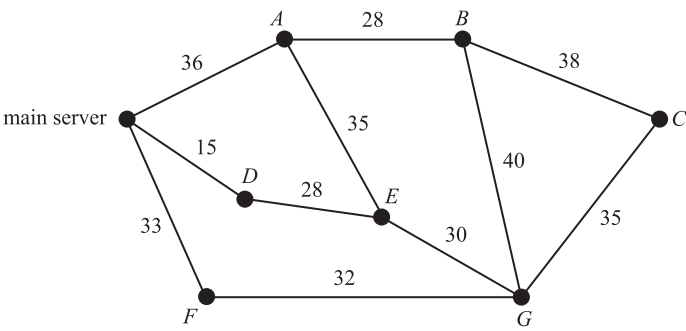
The following network diagram is drawn to represent this information and a correct minimum spanning tree is shown by the solid lines.



Complete the network diagram including a possible value for each of the two edges **AC** and **BE**, and justify why **AC** and **BE** were not included as part of the minimum spanning tree. (3 marks)

25. Networks, STD2 N2 SM-Bank 34

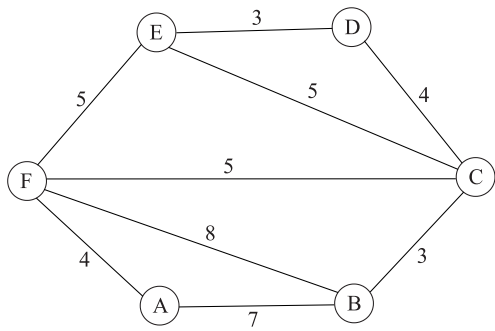
The following diagram shows the distances, in metres, along a series of cables connecting a main server to seven points, **A** to **G**, in a computer network.



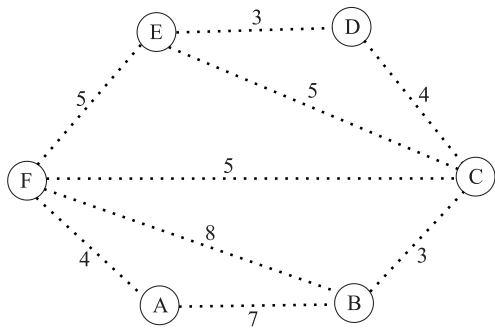
Calculate the minimum length of cable, in metres, required to ensure that each of the seven points is connected to the main server directly or via another point. **(2 marks)**

26. Networks, STD2 N2 SM-Bank 3

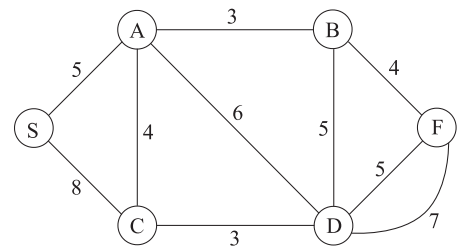
The diagram below is a connected network.



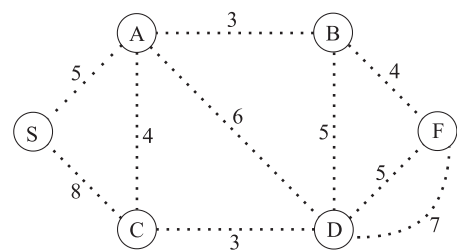
Complete the diagram below to show the minimal spanning tree of this network. **(2 marks)**



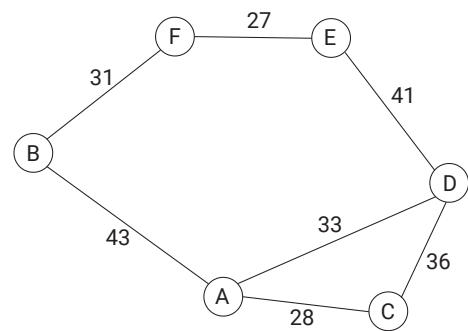
27. Networks, STD2 N2 SM-Bank 4



Complete the minimal spanning tree of the network above on the diagram below. (2 marks)

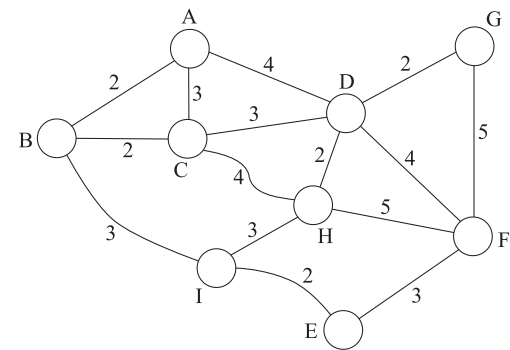


28. Networks, STD2 N2 SM-Bank 8



Highlight the minimal spanning tree of this network on the diagram above, or in a separate diagram. (2 marks)

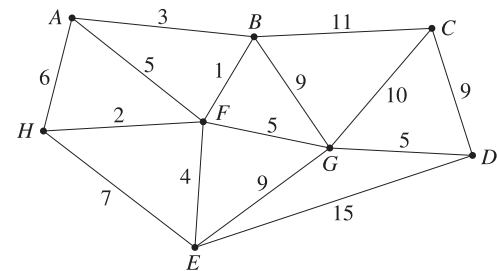
29. Networks, STD2 N2 SM-Bank 9



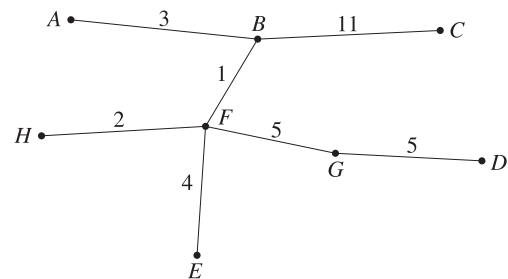
In a separate diagram or on the diagram above, show the minimum spanning tree. (2 marks)

30. Networks, STD2 N2 2023 HSC 19

A network of running tracks connects the points **A, B, C, D, E, F, G, H**, as shown. The number on each edge represents the time, in minutes, that a typical runner should take to run along each track.



- a. Which path could a typical runner take to run from point **A** to point **D** in the shortest time? (2 marks)
- b. A spanning tree of the network above is shown.



Is it a minimum spanning tree? Give a reason for your answer. (2 marks)

Worked Solutions

1. Networks, STD2 N2 2010 FUR1 5 MC

Using Kruskal's algorithm:

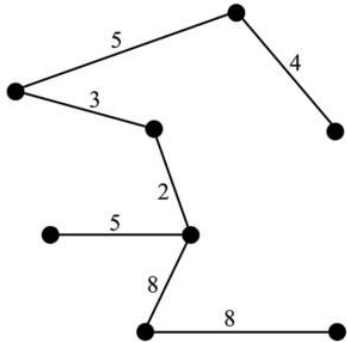
Edge 1: 2 (least weight)

Edge 2: 3

Edge 3: 4

Edges 4-5: 5

Edges 6-7: 8 (unused edges with weights < 8 create circuits and are ignored)



$\therefore$ Minimal spanning tree

$$= 2 + 3 + 4 + 5 + 5 + 8 + 8$$

$$= 35$$

$\Rightarrow C$

2. Networks, STD2 N2 2013 FUR1 3 MC

Using Prim's algorithm

Starting at far left vertex,

1st edge: 2

2nd edge: 3

3rd edge: 4

4th edge: 3 etc...

$\Rightarrow A$

Worked Solutions

3. Networks, STD2 N2 2014 FUR1 5 MC

Using Prim's algorithm:

Starting at vertex A ,

1st edge: $A \rightarrow J$ (6)

2nd edge: $A \rightarrow B$ (8)

3rd edge: $B \rightarrow C$ (9)

4th edge: $J \rightarrow I$ (10)

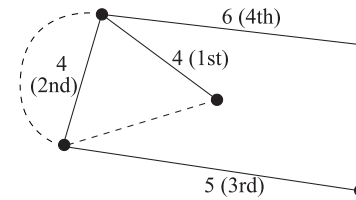
3rd edge: $B \rightarrow D$ (10) etc...

$\Rightarrow A$

4. Networks, STD2 N2 SM-Bank 26 MC

One Strategy – Using Kruskal's Algorithm:

There are 5 vertices, so we need 4 edges.



$\therefore$ Weight = $4 + 4 + 5 + 6$

$$= 19$$

$\Rightarrow B$

5. Networks, STD2 N2 SM-Bank 27 MC

Using Kruskal's Theorem:

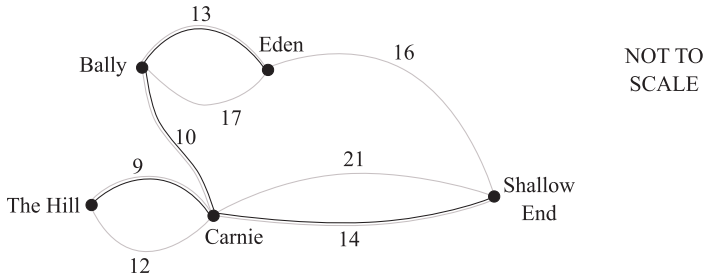
⇒ 5 vertices – 4 edges needed

Edge 1: The Hill → Carnie (9)

Edge 2: Carnie → Bally (10)

Edge 3: Bally → Eden (13)

Edge 4: Carnie → Shallow End (14)

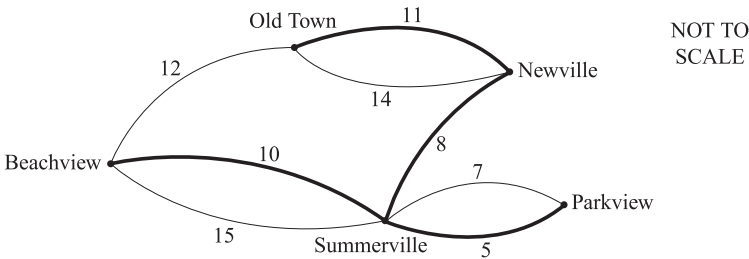


∴ Maximum length = 9 + 10 + 13 + 14
= 46 km

⇒ A

6. Networks, STD2 N2 SM-Bank 1 MC

Using Kruskal's Algorithm:



- 1st edge: Parkview – Summerville (5)
2nd edge: Summerville – Newville (8)
3rd edge: Beachview – Summerville (10)
4th edge: Old Town – Newville (11)

∴ Minimum length of pipes.
= 5 + 8 + 10 + 11
= 34

⇒ B

7. Networks, STD2 N2 2006 FUR1 4 MC

Using Kruskal's Algorithm:

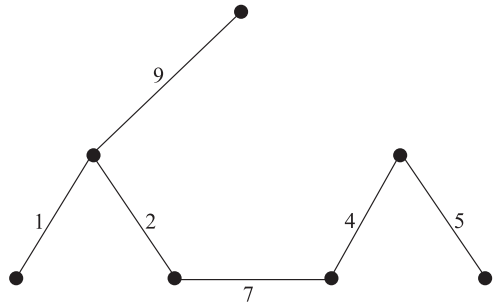
Edge 1: 1 (minimum weight)

Edge 2: 2

Edge 3: 4 (edge weight 3 creates a circuit and is ignored)

Edge 4: 5 etc...

∴ Minimum spanning tree is:

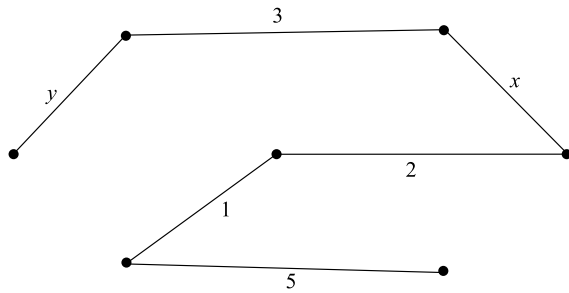


⇒ D

8. Networks, STD2 N2 2007 FUR1 7 MC

Using Kruskal's Algorithm (as a guide),

Minimum spanning tree must be:



$$\therefore 19 = y + 3 + x + 2 + 1 + 5$$

$$8 = x + y$$

$\Rightarrow C$

9. Networks, STD2 N2 2020 FUR1 5 MC

One strategy – Using Prim's Algorithm

Starting at A

1st edge: $AH = 6$

2nd edge: $HG = 5$

then ... $AB = 7$, $GI = 9$, $IE = 6$, $EF = 5$

$DE = 8$, $CD = 7$

Total length = 53 m (not including x)

If $x < 9$, x would replace GI and minimum length would be less than 53m.

$\Rightarrow C$

10. Networks, STD2 N2 2016 FUR1 4 MC

Include k as a given edge before using Kruskal's Algorithm,

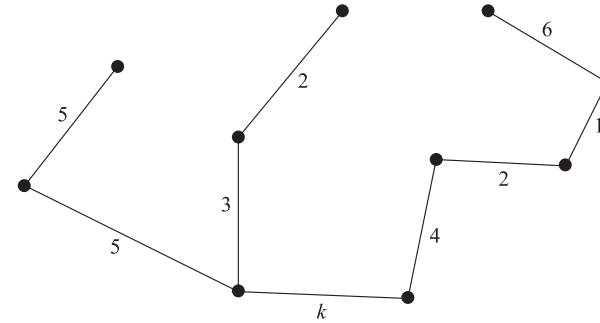
Edge 1: k

Edge 2: 1

Edges 3-4: 2

Edge 4: 3 etc...

$\therefore$ Minimum spanning tree



$$\text{Total weight} = k + 1 + 2 + 2 + 3 + 4 + 5 + 5 + 6$$

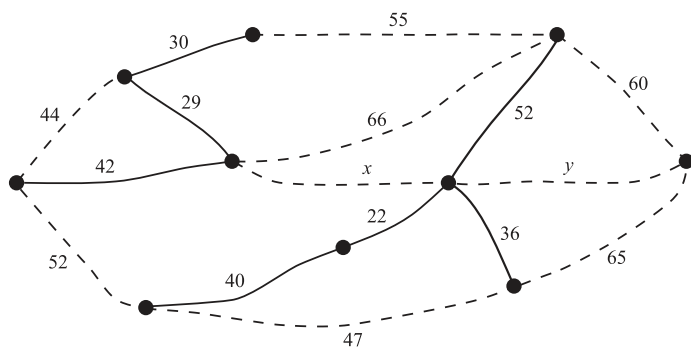
$$33 = k + 28$$

$$\therefore k = 5$$

$\Rightarrow D$

11. Networks, STD2 N2 2021 FUR1 8 MC

A partial minimal spanning tree can be drawn:



Consider each option:

A: If $x = 50$ (include), $y = 55$ (include)

→ Total length = $251 + 50 + 55 \neq 361$ (incorrect)

B: If $x = 50$ (include), $y = 60$ (include)

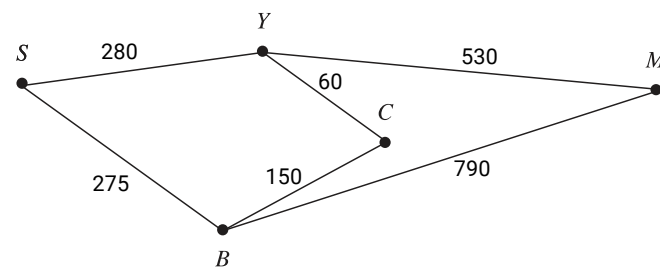
→ Total length = $251 + 50 + 55 = 356$ (correct)

Similarly, options C, D, E can be shown to be incorrect.

⇒ **B**

12. Networks, STD2 N2 2022 HSC 20

a.



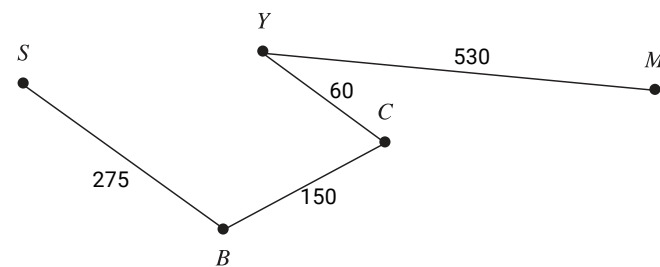
b. Using Prim's algorithm (starting at Y):

1st edge: YC

2nd edge: CB

3rd edge: SB

4th edge: YM



Length of minimum spanning tree

$$= 275 + 150 + 60 + 530$$

$$= 1015 \text{ km}$$

13. Networks, STD2 N2 2015 FUR2 1

a. \$300

b. Minimum cost of K to N

$$= 440 + 480$$

$$= \$920$$

c.i. Using Prim's Algorithm:

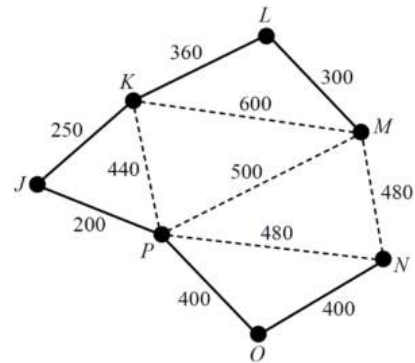
Starting at Vertex L

1st Edge: $L \rightarrow M$ (300)

2nd Edge: $L \rightarrow K$ (360)

3rd Edge: $K \rightarrow J$ (250)

4th Edge: $J \rightarrow P$ (200) etc...



c.ii. Disconnect $J-P$ and $O-P$

$$\text{Savings} = 200 + 400 = \$600$$

Add in $M-N$

$$\text{Cost} = \$480$$

$$\therefore \text{Net savings} = 600 - 480$$

$$= \$120$$

MARKER'S COMMENT: Many students had difficulty finding the minimum spanning tree, often incorrectly excluding PO or KL .

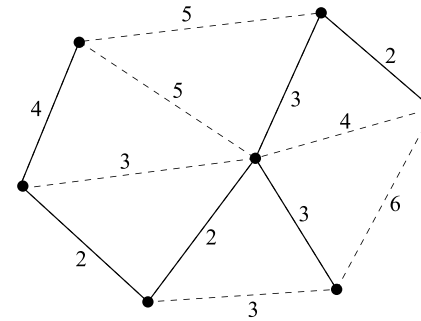
14. Networks, STD2 N2 2008 FUR2 1

a. Using Kruskal's Algorithm

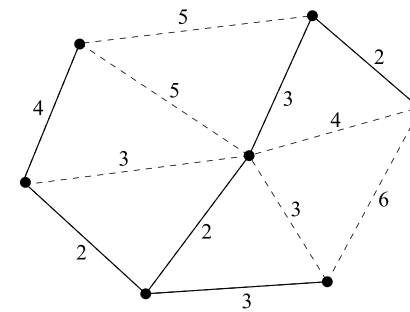
Edges 1-3: 2

Edges 4-5: 3 (2 edges with weight 3 create a circuit and are ignored)

Edge 6: 4



or



b. Length of minimal spanning tree

$$= 2 + 2 + 2 + 3 + 3 + 4$$

$$= 16 \text{ metres}$$

c. 2

15. Networks, STD2 N2 2011 FUR2 2

a. Using Prim's Algorithm

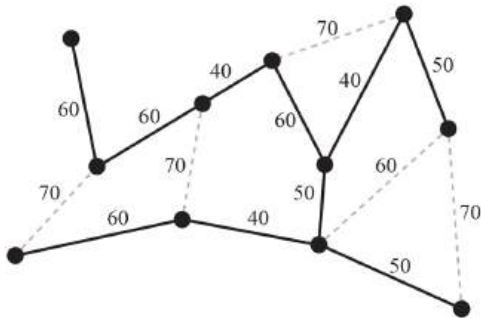
Starting at bottom right vertex

1st Edge: 50

2nd Edge: 40

3rd Edge: 50

4th Edge: 40 etc...



b. Total length of water pipe

$$= 50 + 40 + 50 + 40 + 50 + 60 + 40 + 60 + 60 + 60$$

$$= 510 \text{ metres}$$

16. Networks, STD2 N2 SM-Bank 2

i. One Strategy: Using Prim's Algorithm

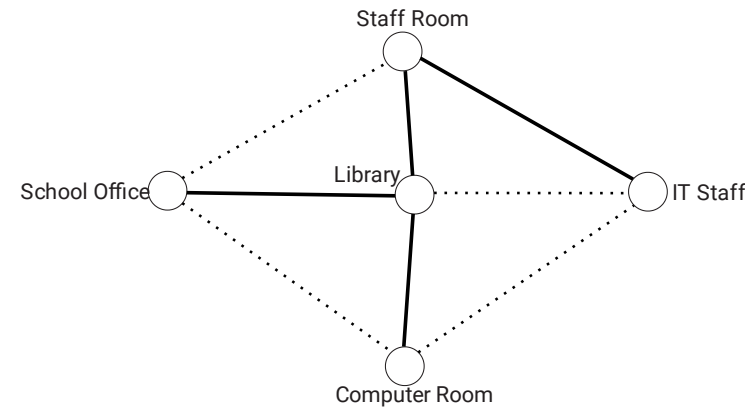
Starting vertex – Staff Room

1st edge: Staff Room – Library

2nd edge: Library – School Office

3rd edge: Staff Room – IT Staff

4th edge: Library – Computer Room



$$\text{ii. Minimum Cost} = 300 + 300 + 400 + 500$$

$$= \$1500$$

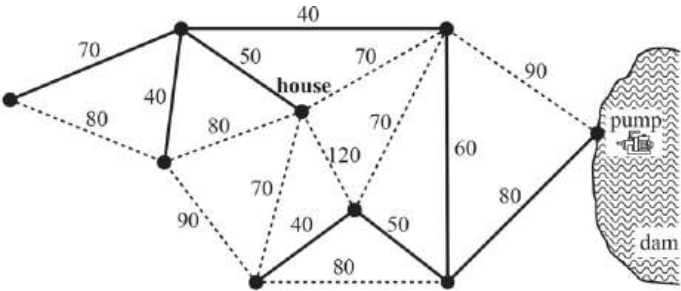
17. Networks, STD2 N2 SM-Bank 14

i. 2 (the house and the top right vertex)

ii. One Strategy – Using Prim’s algorithm:

- Starting at the house
- 1st edge: 50
- 2nd edge: 40 (either)
- 3rd edge: 40
- 4th edge: 60 etc. . .

MARKER’S COMMENT: Many students, surprisingly, had problems with part (i).

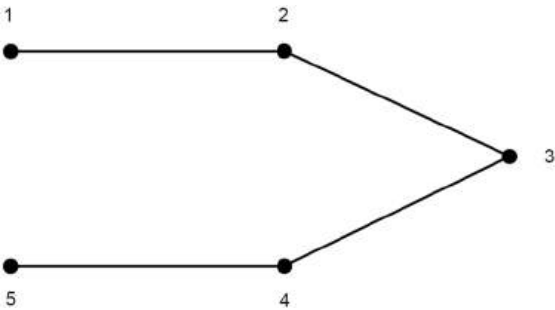


iii. Minimum spanning tree

18. Networks, STD2 N2 2007 FUR2 1

a. Minimum number of edges = 4

b. One of many possibilities,

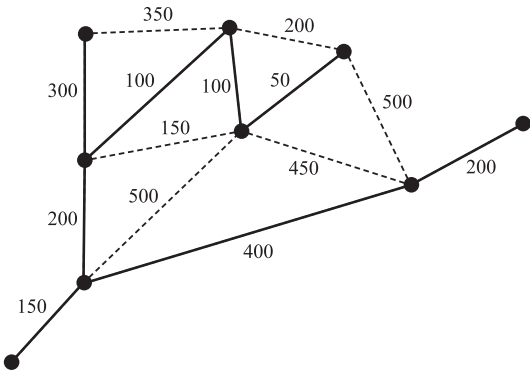


19. Networks, STD2 N2 2017 FUR2 3a

a. Minimum spanning tree.

b. Using Kruskal’s Algorithm

- Edge 1: 50
- Edges 2–3: 100
- Edge 4: 150 (ignore the 150 edge that creates a circuit)
- Edges 5–6: 200 (ignore the 200 edge that creates a circuit)
- Edge 7: 300
- Edge 8: 400



20. Networks, STD2 N2 2019 HSC 30

a. One strategy – using Prim's algorithm:

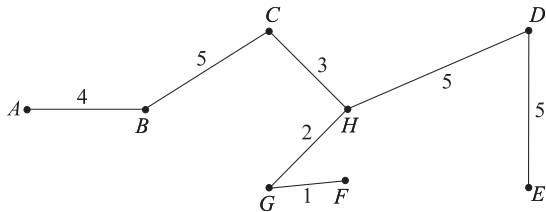
Starting at *A*

1st edge - *AB*, 2nd edge - *BC*

3rd edge - *CH*, 4th edge - *HG*

5th edge - *GF*, 6th edge - *HD*

7th edge - *DE* or *HE*



Maximum length = $4 + 5 + 3 + 2 + 1 + 5 + 5 = 25$ km

b. Shortest Path is *CGHE*

21. Networks, STD2 N2 2020 HSC 18

a. Using Kruskal's Algorithm (one of many possibilities):

Edge 1 : *GH* (1)

Edge 2 : *FH* (2)

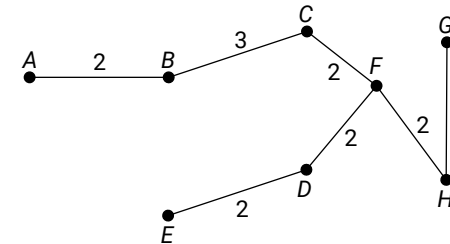
Edge 3 : *CF* (2)

Edge 4 : *FD* (2)

Edge 5 : *DE* (2)

Edge 6 : *BC* (3)

Edge 7 : *AB* (2)



Minimum length of spanning tree = $1 + 2 + 2 + 2 + 2 + 3 + 2$
= 14

b. Add *CK* to the minimum spanning tree in (a).

$\therefore$ Revised length = $14 + 10$
= 24

22. Networks, STD2 N2 2021 HSC 23

a. Using Prim's algorithm (starting at *W*):

1st edge: *WM*

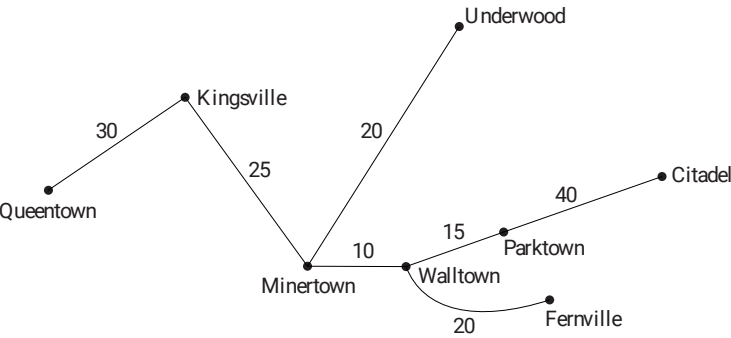
2nd edge: *WP*

3rd/4th edges: *MU* and *WF*

5th edge: *MK*

6th edge: *KQ*

7th edge: *PC*

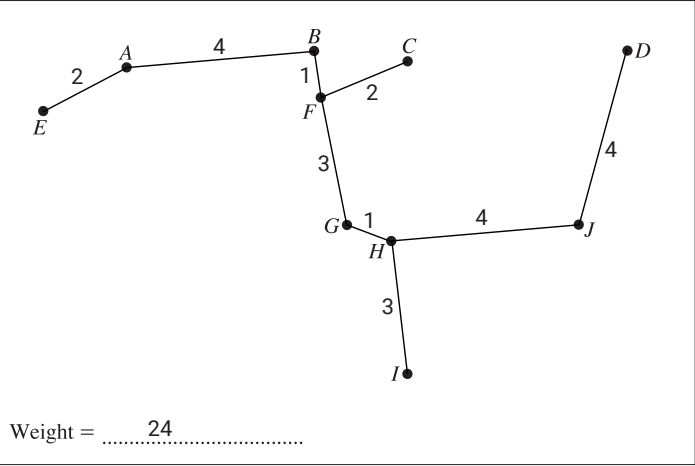


Length of minimum spanning tree = 160

b. Fastest route (*Q* to *U*) = 45 + 20
= 65 minutes

23. Networks, STD2 N2 2024 HSC 18

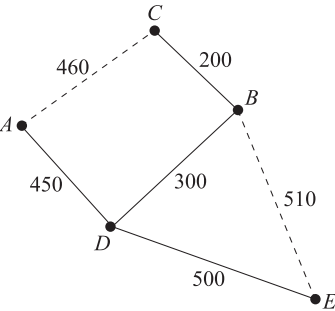
a.



b. Yes.

→ The edge *FC* on the MST above could be replaced by the edge *BC* to create a second MST (with ex

24. Networks, STD2 N2 SM-Bank 30



Consider Kruskal's algorithm for the given minimum spanning tree:

CB → 1st edge (200 – minimum weight)

BD → 2nd edge (300)

AD → 3rd edge (450)

⇒ *AC* = 460 (choose any value such that *AC* > 450)

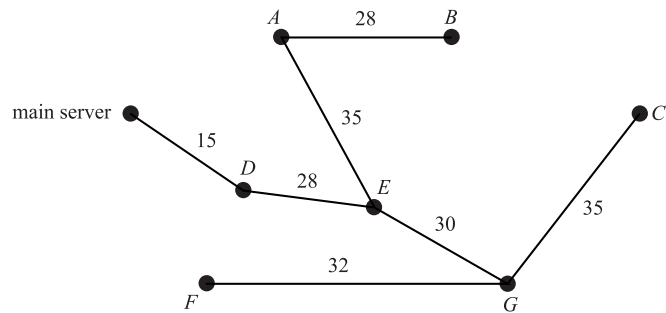
DE → 4th edge (500)

⇒ *BE* = 510 (choose any value such that *BE* > 500)

∴ *BE* > 500 (*BE* must be greater than *DE*)

∴ *AC* > 450 (*AC* must be greater than *AD*)

25. Networks, STD2 N2 SM-Bank 34



Using Prim's Algorithm:

Edge 1: main server to D (15)

Edge 2: DE (28)

Edge 3: EG (30)

Edge 4: GF (32)

Edge 5: EA (35)

Edge 6: AB (28)

Edge 7: GC (35)

$\therefore$ Minimum length

$$= 15 + 28 + 30 + 32 + 35 + 28 + 35$$

$$= 203 \text{ m}$$

26. Networks, STD2 N2 SM-Bank 3

One Strategy – Kruskal's algorithm

There are 6 vertices, so we need 5 edges.

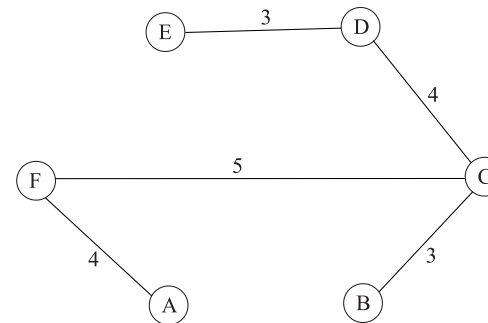
Edge 1: BC (least weight) – could have chosen ED

Edge 2: DE (next least weight)

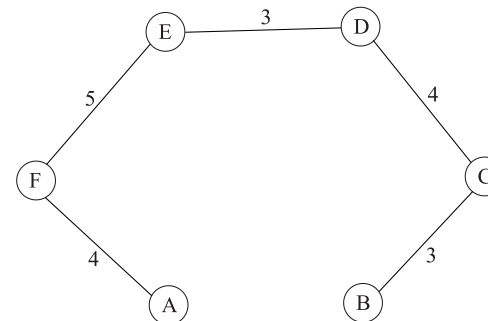
Edge 3: AF (could have been CD)

Edge 4: CD

Edge 5: CF or EF (reject CE as it creates a cycle)



or



27. Networks, STD2 N2 SM-Bank 4

One Strategy: Using Prim's algorithm

6 vertices → need 5 edges.

Starting at vertex A (any vertex can be chosen)

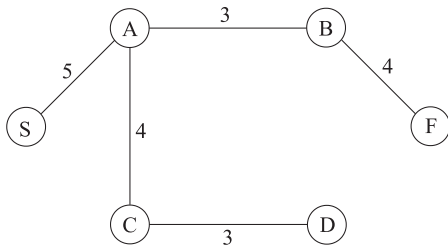
1st Edge: AB

2nd Edge: AC (BF also possible)

3rd Edge: CD

4th Edge: BF

5th Edge: AS



28. Networks, STD2 N2 SM-Bank 8

One Strategy – Using Prim's algorithm:

6 vertices → 5 edges on spanning tree

Starting at vertex A

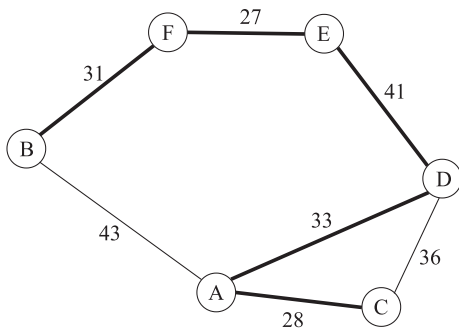
Edge 1: AC

Edge 2: AD

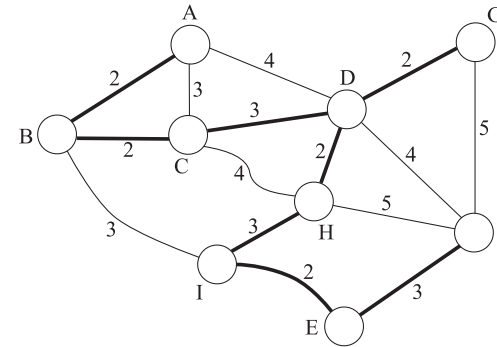
Edge 3: ED (reject CD which creates a cycle)

Edge 4: EF

Edge 5: BF



29. Networks, STD2 N2 SM-Bank 9



One strategy – Using Kruskal's algorithm:

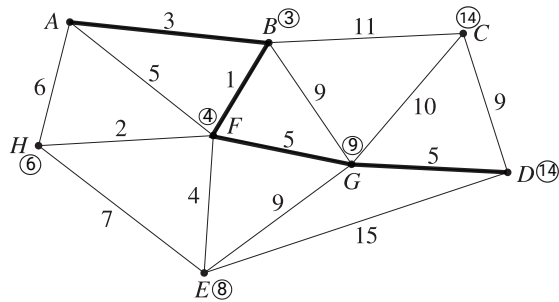
Edges 1 – 5: AB, BC, DG, DH and EI

Edges 6 – 8: CD, EF and HI

(note AC cannot be chosen → creates a cycle)

NB: There is more than one minimal spanning tree in this circuit (having a weight of 19).

a. Using Dijkstra's Algorithm:



Shortest route = **ABFGD**

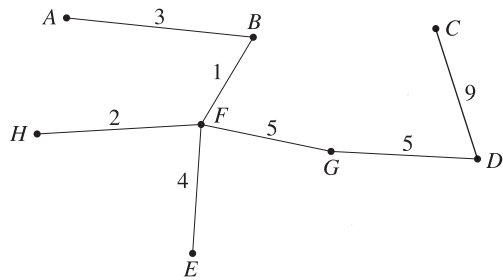
$$\begin{aligned} &= 3 + 1 + 5 + 5 \\ &= 14 \end{aligned}$$

b. Total time of given spanning tree

$$\begin{aligned} &= 3 + 11 + 1 + 2 + 4 + 5 + 5 \\ &= 31 \end{aligned}$$

♦ Mean mark (b) 47%.

Consider the MST below:



Total time (MST) = 3 + 1 + 2 + 4 + 5 + 5 + 9 = 29
∴ Given tree is NOT a MST.